

PURE Workshop 2026

Introduction to Data Visualization and Statistics



POLYPLOIDY
INTEGRATION
& INNOVATION
INSTITUTE



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Why do we start here?



Your closest collaborator is you six months ago, and you do not reply to emails -Karl Broman



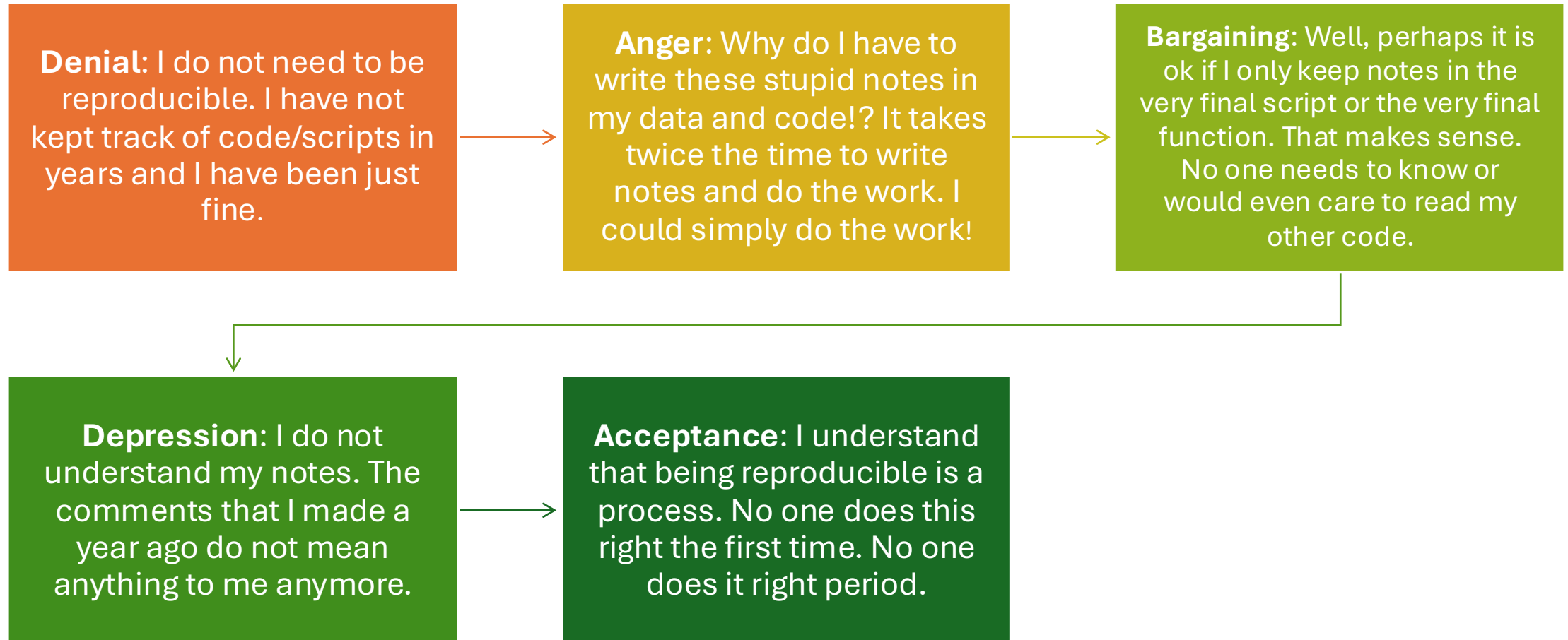
The most important tool is the mindset, when starting, that the end product will be reproducible - Keith Baggerly



Assume that everything that you are doing right now will need to be redone at some point in the future: be prepared

Five stages of reproducibility

by Claudia Solis-Lemus (UW Maddison)



Ways to keep organized your work

>  begoniaceae_shared	✓	May 28, 2026 at 8:26 AM	--	Folder
>  figures	✓	May 28, 2026 at 8:20 AM	--	Folder
>  revresults	✓	May 28, 2026 at 8:09 AM	--	Folder
>  revcode	✓	May 27, 2026 at 10:38 AM	--	Folder
>  revdata	✓	May 27, 2026 at 10:18 AM	--	Folder
>  original_data	✓	May 14, 2026 at 10:25 AM	--	Folder

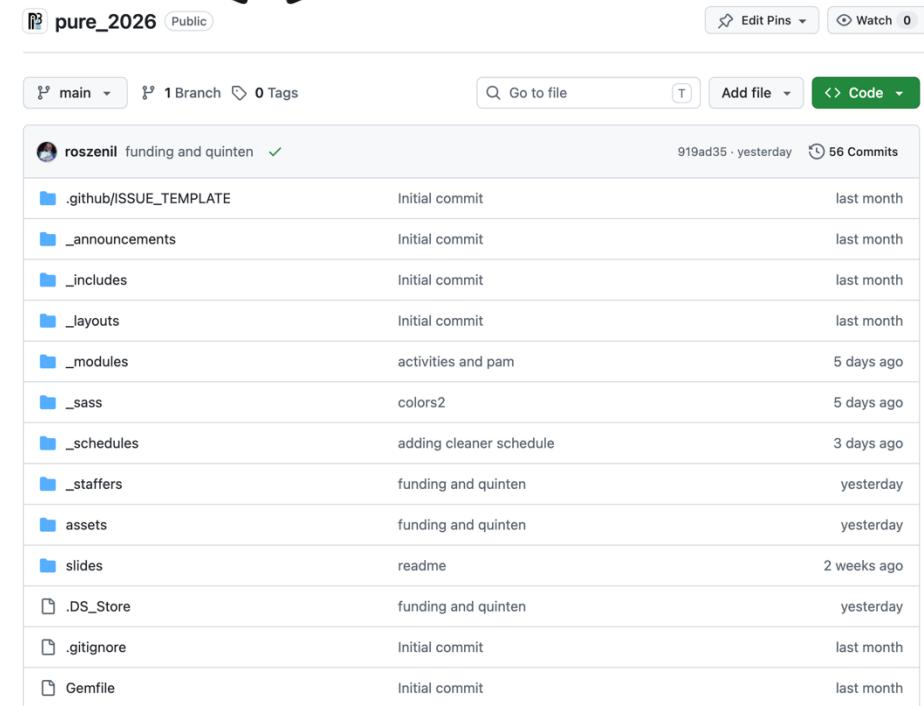


Cloud storage systems

- Backup in case your computer dies
- Automatic
- Private

Code version platforms

- Keep track of file changes
- Collaboration
- Big learning curve and aimed for big projects



R Markdown and Rstudio

Reproducible notebook for statistics and figures



.Rmd files • Develop your code and ideas side-by-side in a single document. Run code as individual chunks or as an entire document.

Dynamic Documents • Knit together plots, tables, and results with narrative text. Render to a variety of formats like HTML, PDF, MS Word, or MS Powerpoint.

Reproducible Research • Upload, link to, or attach your report to share. Anyone can read or run your code to reproduce your work.

What we will be working today

01

Manipulating a dataset using a computer without modifying the original data

02

Making bar plots using a visualization tool called ggplot2

03

Making five-number summary statistics and plots for descriptive statistics

Background and dataset



Porturas et al. (2019) conducted literature surveys that integrated floral morphology, flowering phenology, and reproductive output with neopolyploidy.



We will compare **flower diameter** in neopolyploids and their parents.



Testing **the gigas effect** = Polyploids have greater quantities of DNA and this causes larger cells and translates into larger tissues and organs

Ask questions!

Join at
slido.com
#4108 804



Type your question 🔍 😊 160

Question AI

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R Rosana ZF ▾ Send

If you want us to go to your code in Posit.cloud please put your name (or if you want us to know you!)

Ask questions without judgement (just be polite)

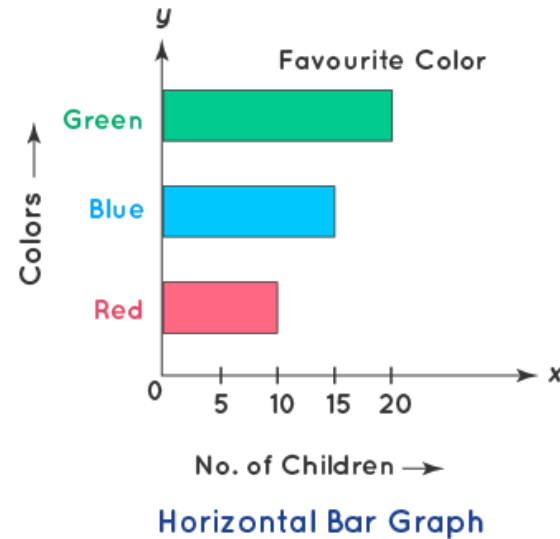
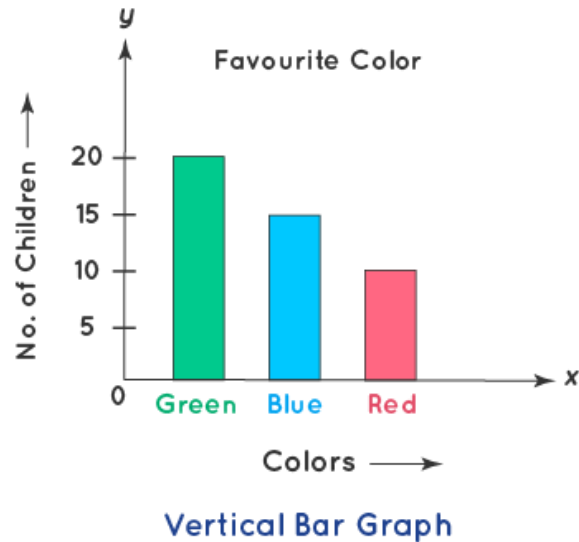
Type your question 🔍 😊 160

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Datasets are complicated



A bar diagram is the graphical representation of precise data using rectangular bars with equal gaps between them.

	Category	Total
This example	Colors	Children
Our example- RStudio	Traits	Studies

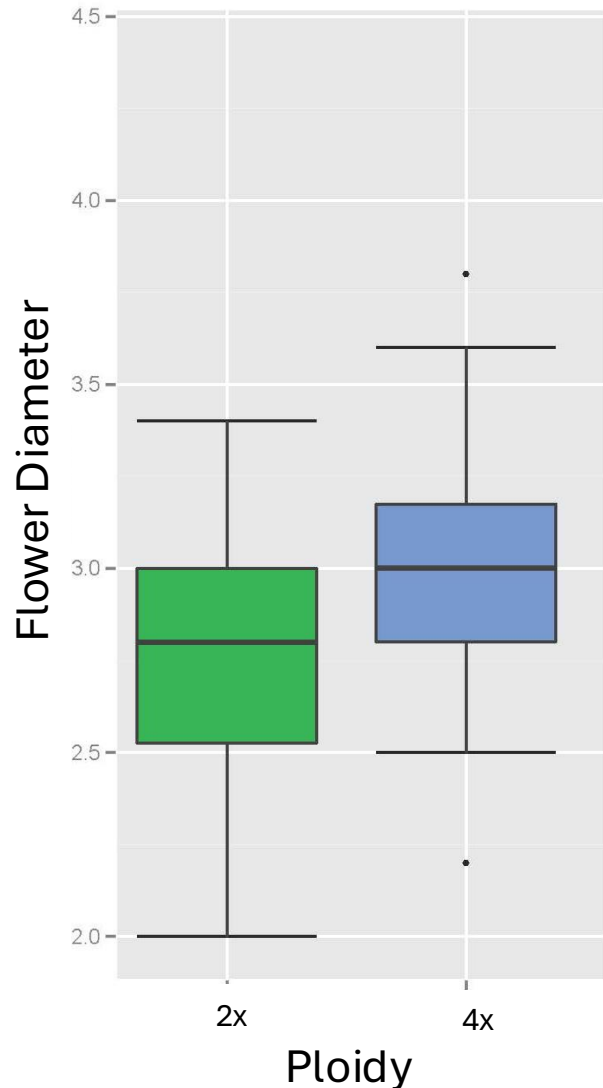
Choose one trait: Flower diameter



Are the flowers of neopolyploids bigger?

- Studies induced polyploidy using colchicine
- Measure diameter before treatment (2x) and after treatment (4x)

Flower diameter. Five-Point Summary



Box and Whisker plots

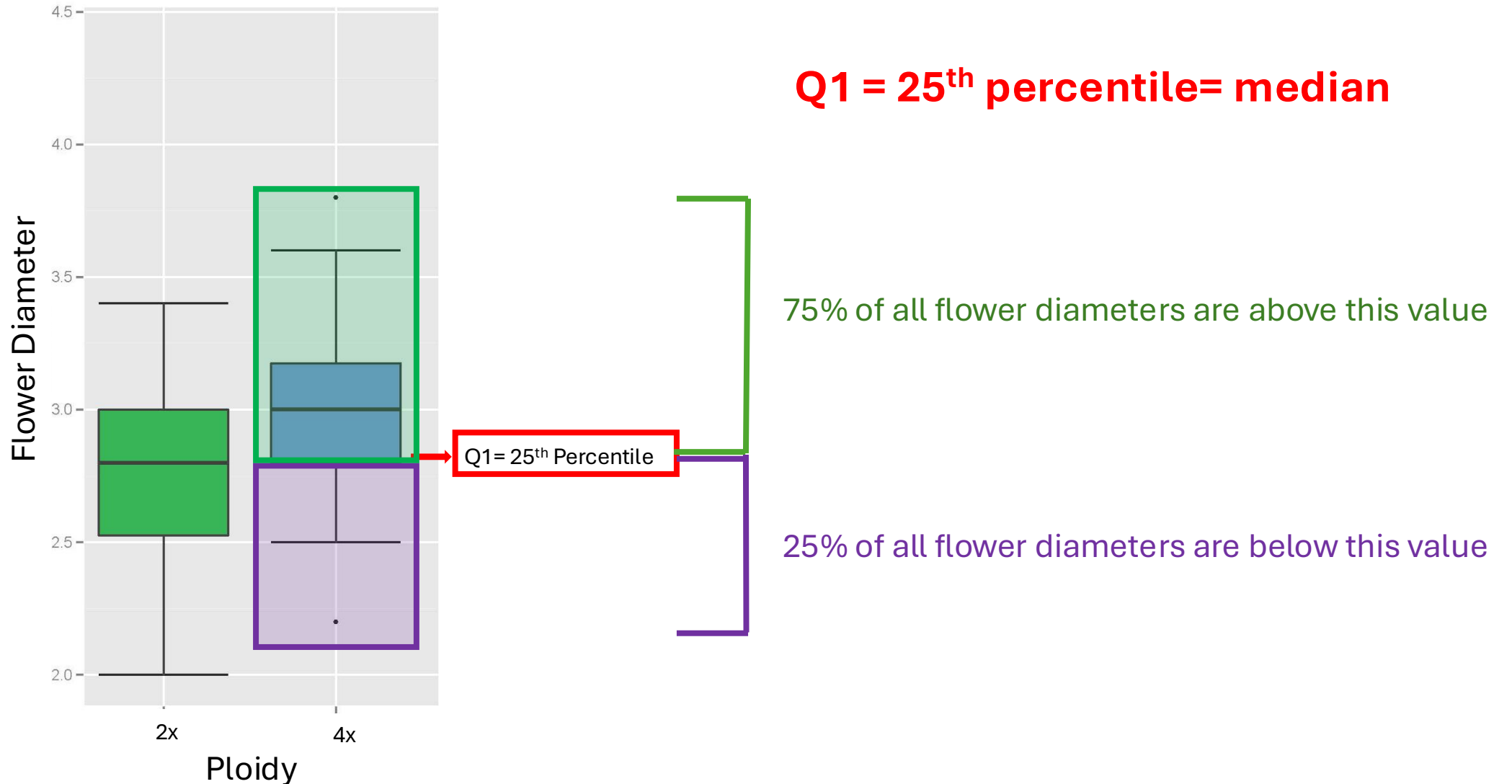
A box and whisker plot is defined as a graphical method of displaying variation in a set of data

Why do we need a visual?

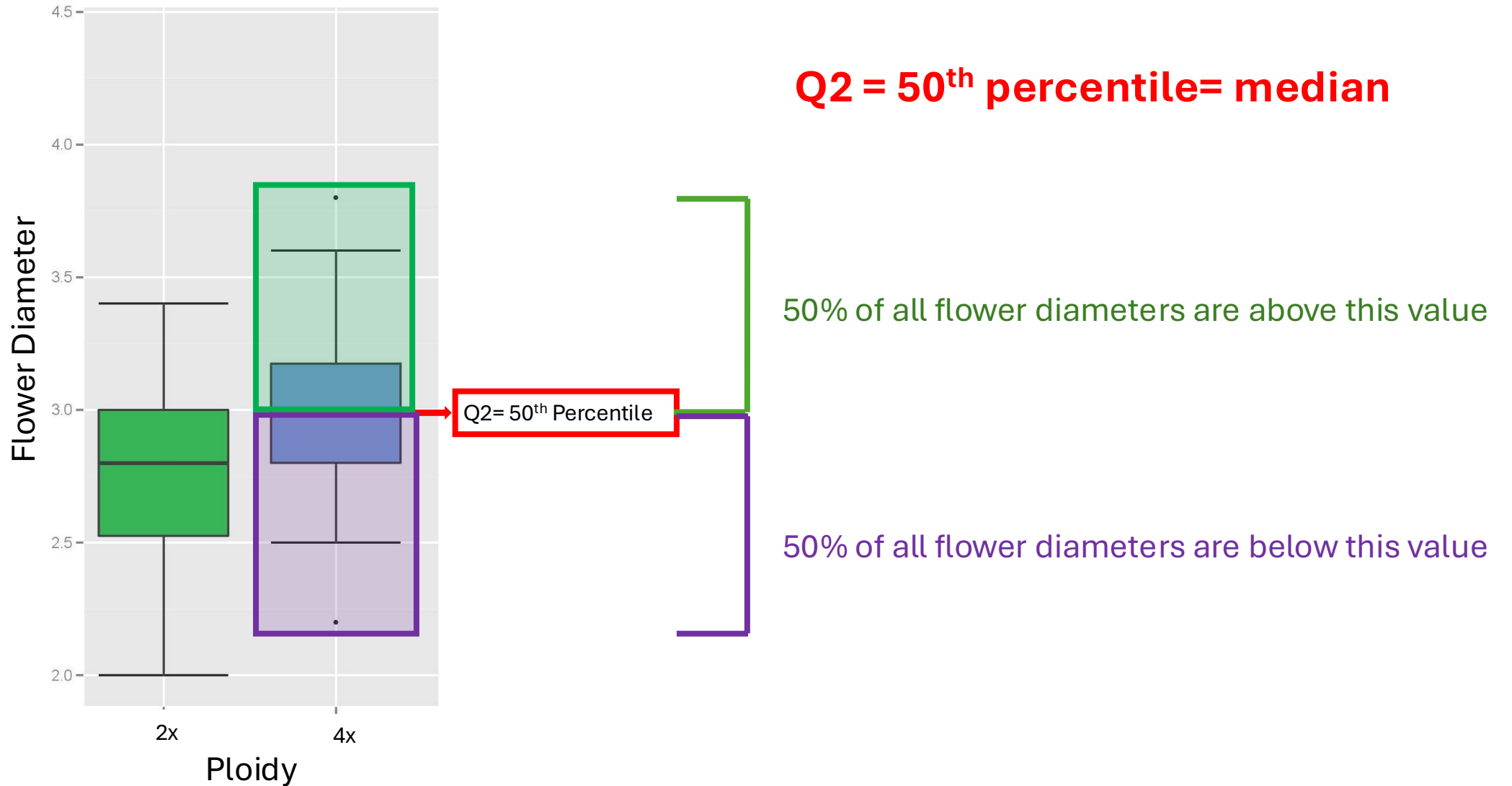
An image is worth more than 100 words

We do not know if the distribution of flower diameters for neo-tetraploids is similar or has the same spread as the diploids

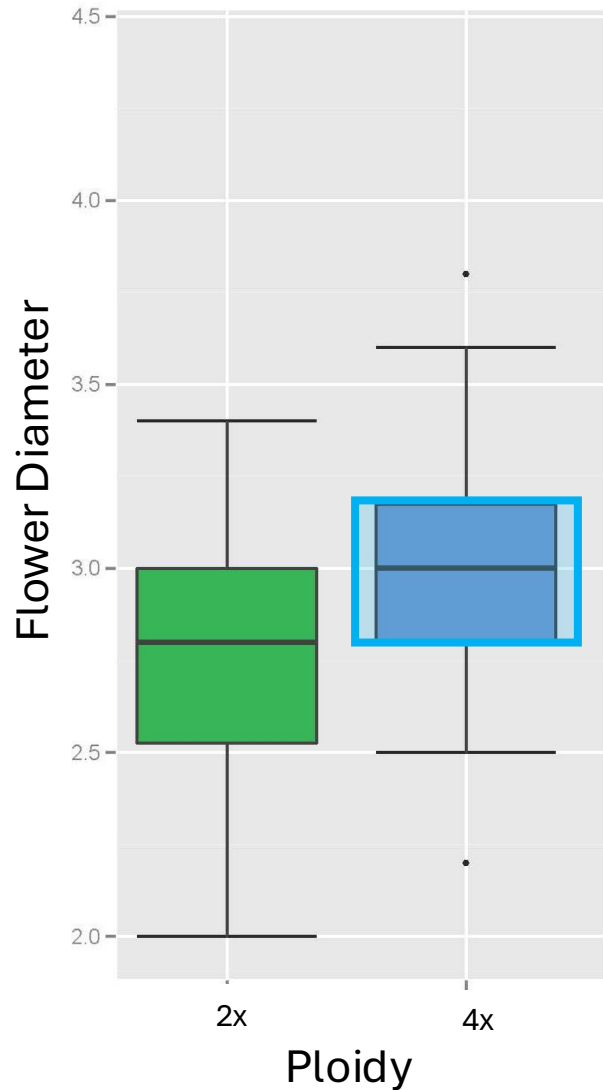
Flower diameter. Five-Point Summary



Flower diameter. Five-Point Summary



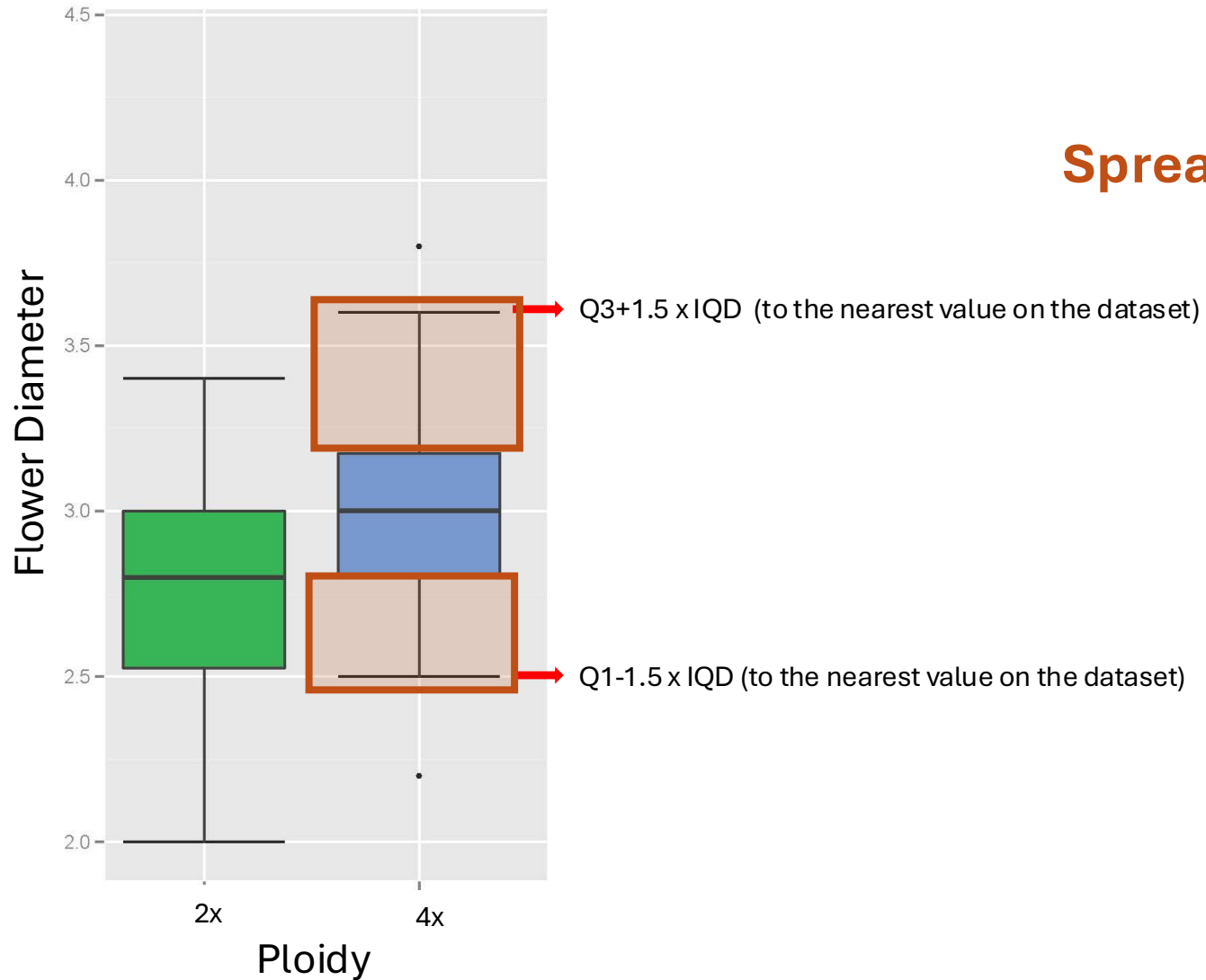
Flower diameter. Five-Point Summary



Interquartile Distance = IQD
= $Q3 - Q1$

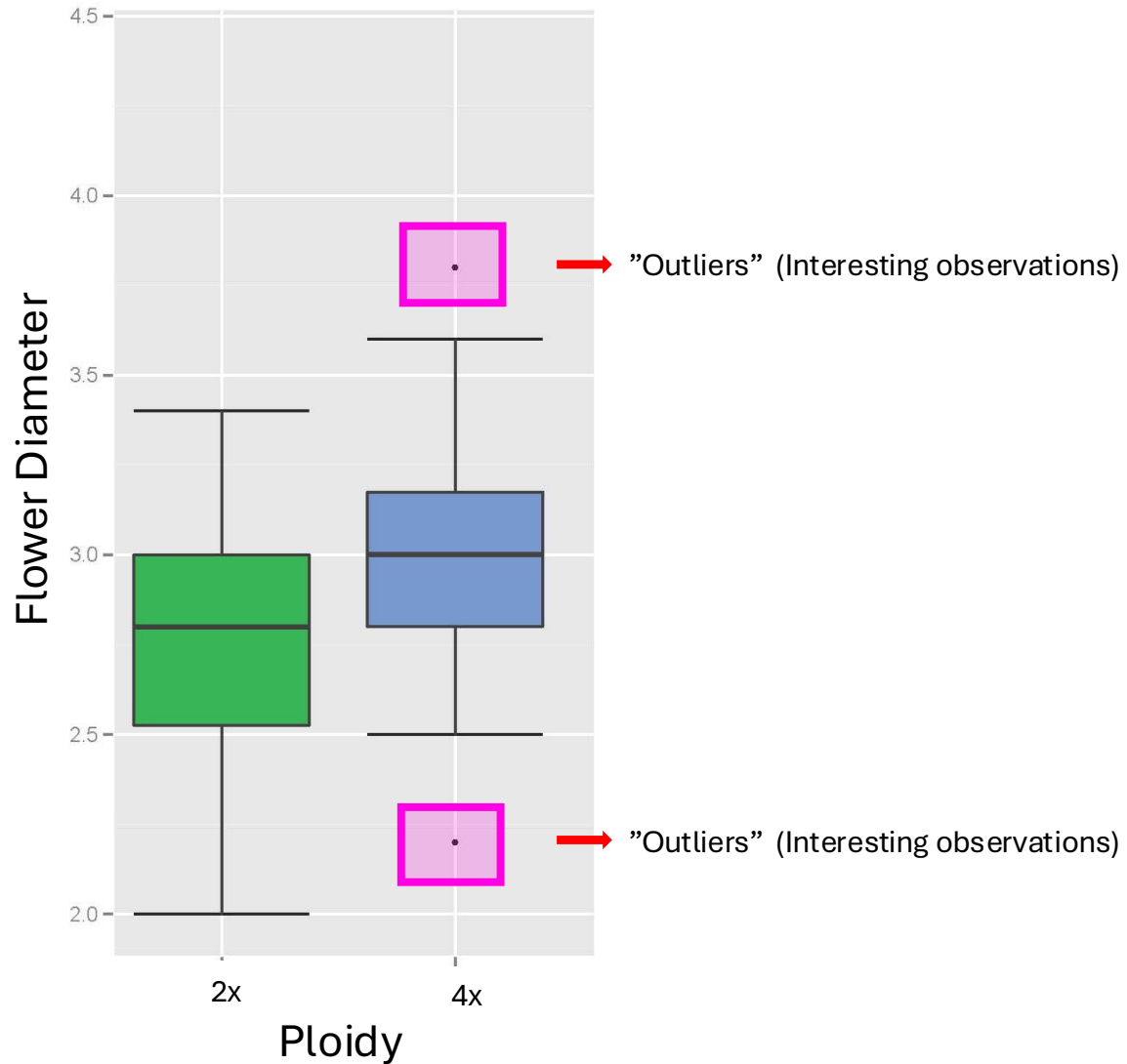
Interquartile Distance
IQD
Are central values similar?

Flower diameter. Five-Point Summary



Whiskers
Spread of the distribution

Flower diameter. Five-Point Summary



Outliers
Interesting values worth
exploring

Information and links



[↗](#) Workshop activities

Monday

06/29/2026:

INTRODUCTION

[Slides](#)

Doug Soltis and Pam Soltis. University of Florida

INTRODUCTION TO DATA VISUALIZATION

[Setting up](#)

Activity

[Link](#)

Rosana Zenil-Ferguson and Pamela Rueda Cediell. University of Kentucky

INTRODUCTION TO PROTEOMICS

[Link](#)

Sixue Chen. University of Mississippi

Ask questions!

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<https://posit.cloud/>

```
report.Rmd x
Go to file/function
Addins
Source Visual
Outline
1 ---
2 title: "Document Title"
3 author: "Author Name"
4 output:
5   html_document:
6     toc: true
7 ---
8
9 ```{r setup, include=FALSE}
10 knitr::opts_chunk$set(echo = TRUE)
11 ```
12
13 ## R Markdown
14
15 This is an R Markdown document.
16 Markdown is a simple formatting
17 syntax for authoring HTML, PDF,
18 and MS Word documents.
19
20 ```{r cars}
21 summary(cars)
22 ```
R Markdown Including Plots
1:1 # Document Title R Markdown
Console Terminal x Render x Background Jobs x
.../rmarkdown/report.Rmd
Output created: report.html
```